Recap

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Recap

- Employed methods like PCA and dimensionality reduction
- Learned about classifiers and regression models
- Worked with datasets and open dataset

Dimensionality reduction

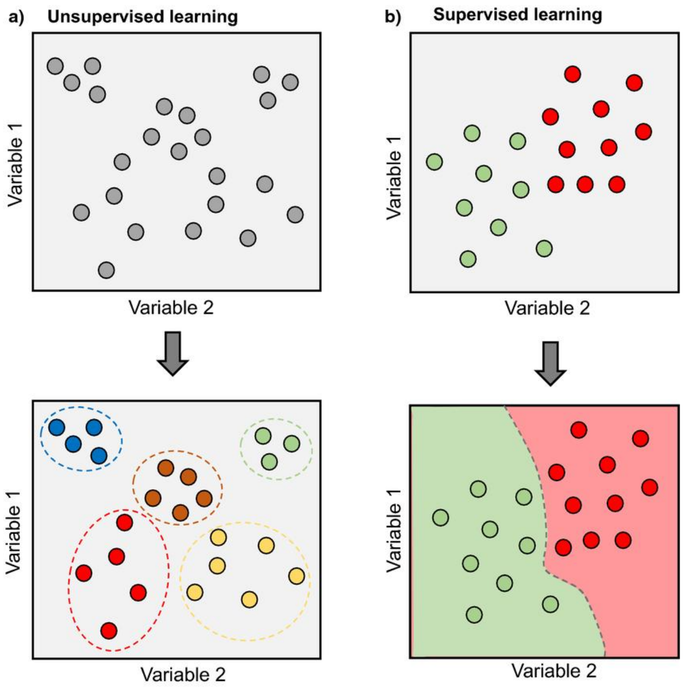
Method	*Type*	*Output*	*Best for*
PCA	Linear	Axes of variance	Denoising, compression
t-SNE	Nonlinear	3d/2d embeddings	Visualisation of local clusters
UMAP	Nonlinear	3d/2d embeddings	Visualisation + structural relationships
VAE	Deep learning, nonlinear	Low-dim latent space + decoder	Modelling, simulation, generative tasks

Classification models

- Learn to categorize data into distinct groups or classes

- supervised classification:
    - Support vector machines: linearly separable data
    - Random forest: non-linearly separable data
- unsupervised clustering:
    - K-means: clustering of similar instances

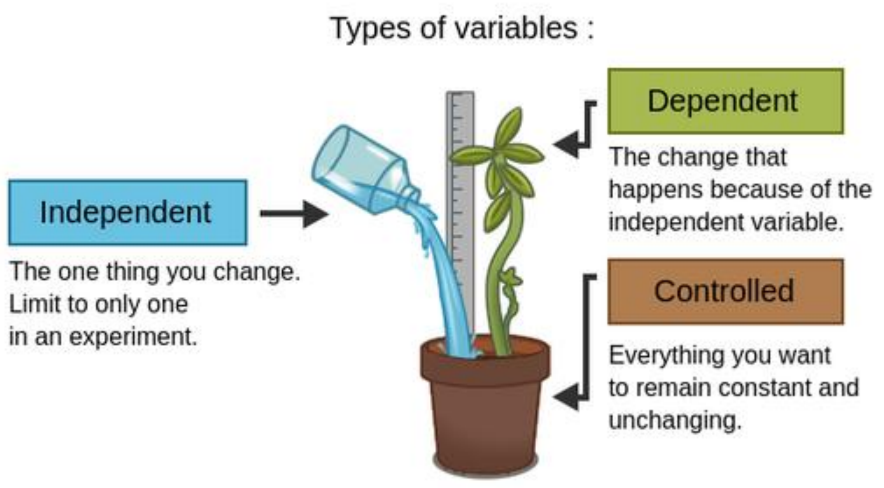
- Artificial neural networks: wide range of problems



Regression

- Learn statistical trends in the data, predict one variable based on others

- Linear regression
- Multi-linear regression
- Logistic regression/GLMs



Contents

- Fail cases of machine learning
- How we can be robust
- Data augmentation

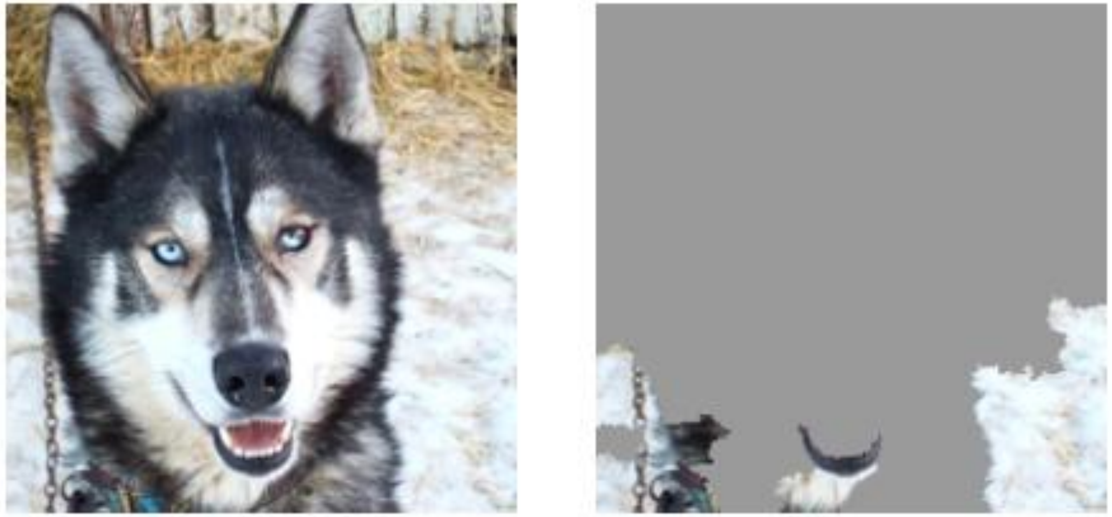
What does accuracy tell us?

How much of **this data** we got correct

But does our dataset represent everything?

Example

- A model trained to distinguish between dogs and wolfs



(a) Husky classified as wolf (b) Explanation

Jiawei, "One pixel attack for fooling deep neural networks." *IEEE TEC* 2019
Ribeiro, Singh, and Guestrin. "" Why should i trust you?" Explaining the predictions of any classifier." ACM SIGKDD 2016.

Investigating robustness

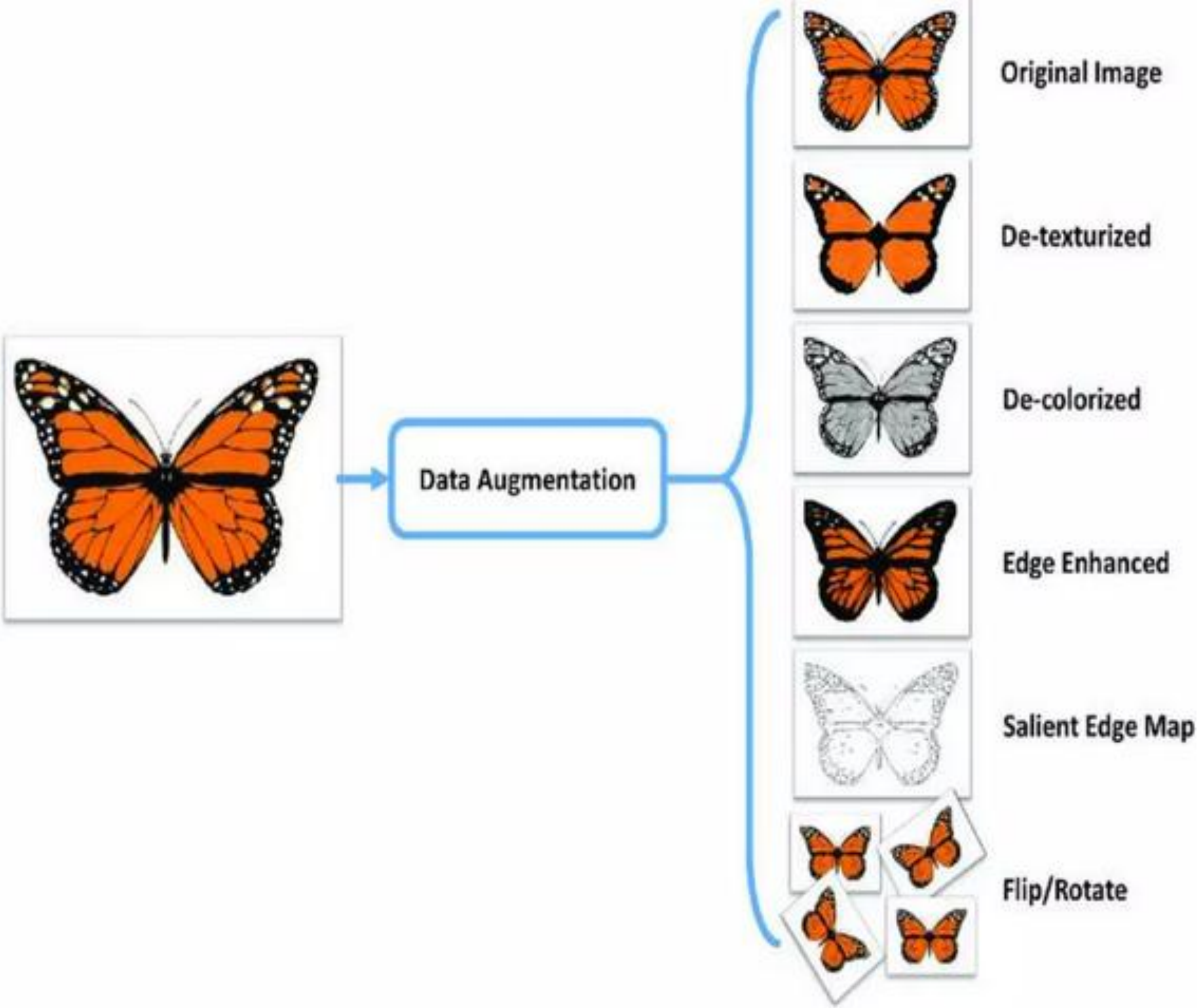
- Test the data on unseen data!
- Still this might not be enough, lets say it is based on weather data, if you only gathered in one season it will almost certainly get incorrect predictions – how can we solve?
- Data augmentation – what exactly has the model learned?

Data augmentation

- Changing your original data to be nosier, or less clear
- Can improve model accuracy when working with small datasets

Data Augmentation

When we have a small dataset of images, we can **oversample** them by performing **data augmentation**.



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What augmentation should I use?

This depends on your *data* and *task*, as some augmentation may negatively impact your model

What types of data modifications preserve the original meaning for your task?

The key thing to remember

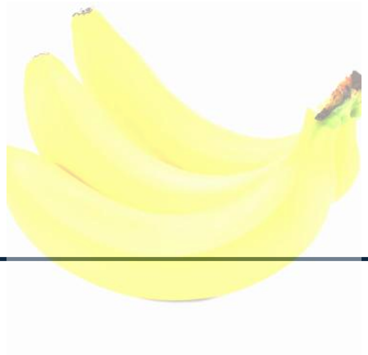
- Your data still must represent the trends!

Banana detection

Detexturized

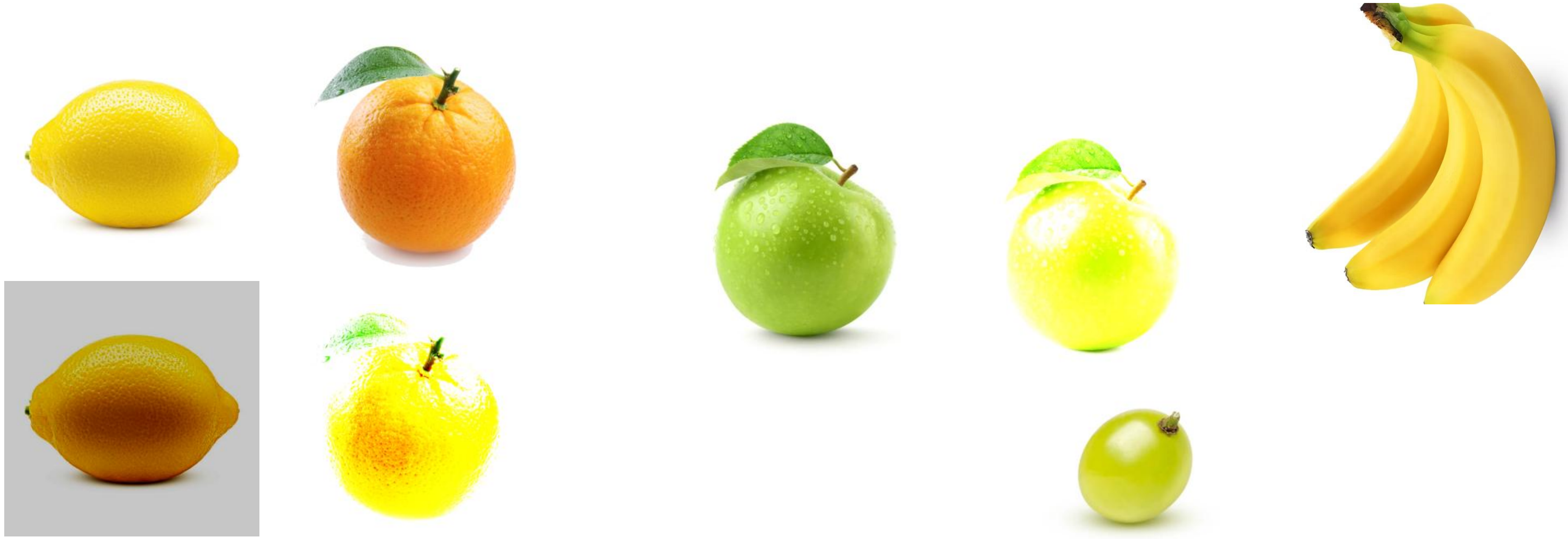
Edge enhance

Flip and rotate



A larger classifier of fruit

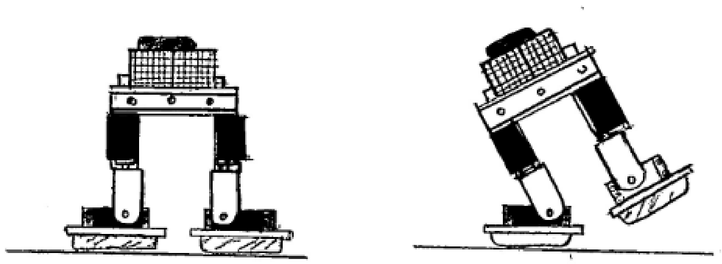
Maybe the lighting does matter...

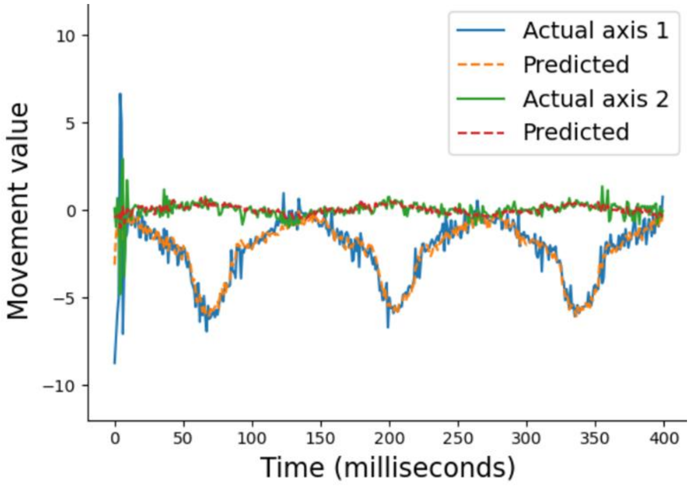


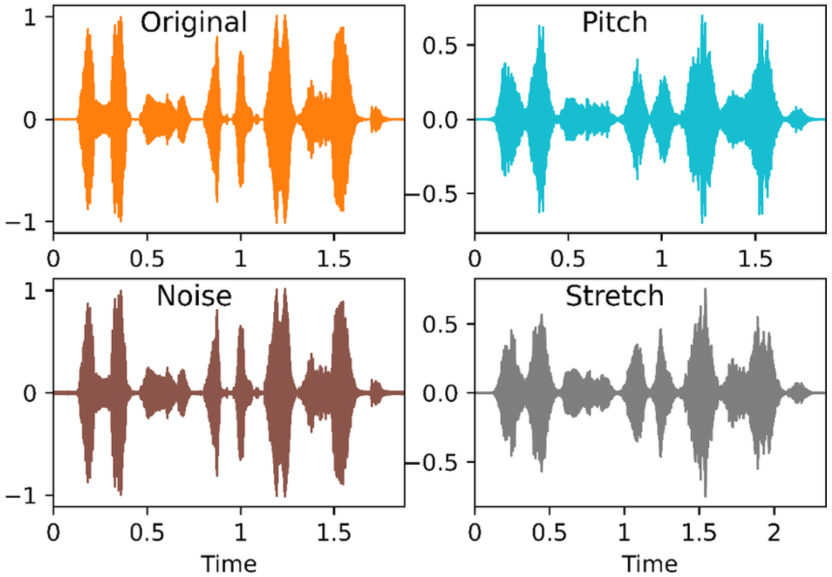
Signals

Augmentation is not limited to images…

Make sure to evaluate what data is relevant







Signals

Audio signals “hello world”

Stretch it: makes it slower spoken “heeellloooooo wwwoorrlldddd”

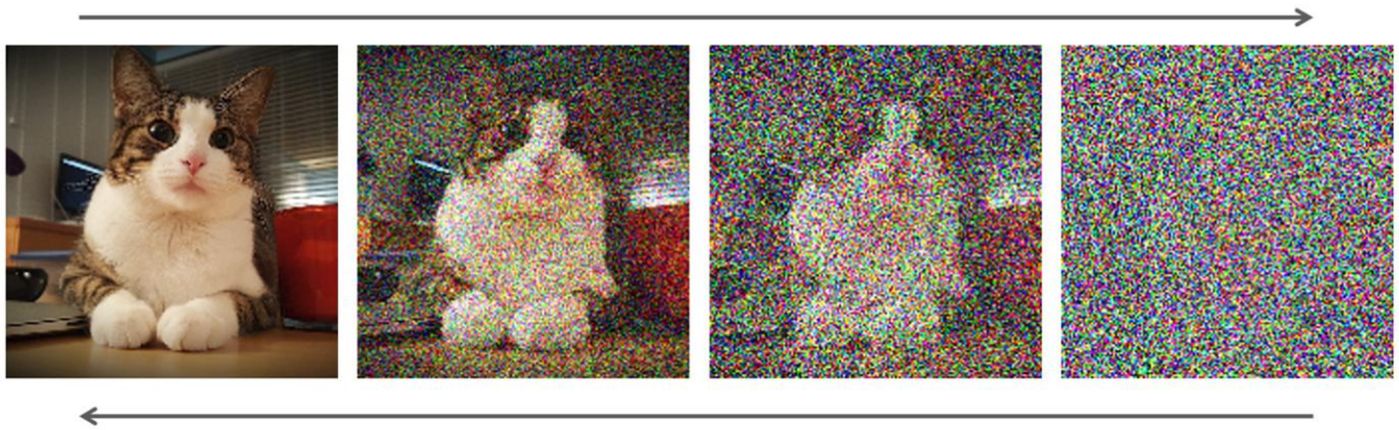
Change the pitch augments it to different voices

This could be helpful for voice recognition

Adding noise

Can you rely on good image quality all the time?

Adding noise helps a model understand beyond specific colours



AI generated datasets

There is an approach of using AI to generate data.

Tasks where a lot of data exists such as image generation, this can be an inexpensive way of generating data

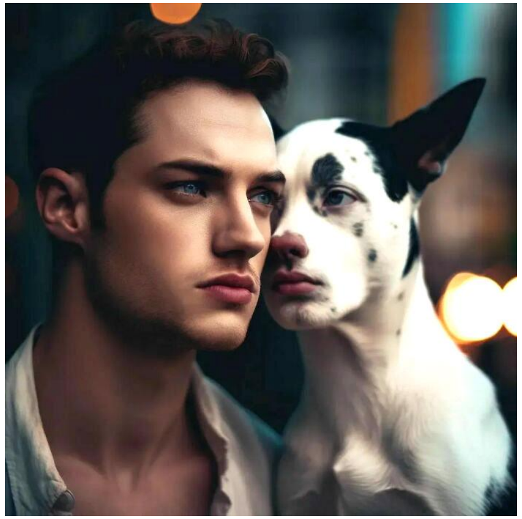
Generative AI can alter your existing images so there is a mix of truth and changes

Problems

Bias

Data not being real enough



But they have some uses

Maybe not best for a full dataset… but can be used for transfer learning

Today

- Catch up on worksheets
- Go through worksheets with a new dataset from kaggle
- Start implementing data augmentation on some of the models you have already developed
    - Eg. Add noise (investigate how much noise helps, and what too much noise looks like), feel free to play around with AI generated data from ChatGPT and see if this actually helps

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